Statistical Methods I: Linear Model

Lec 07: Multiple Linear Regression - Introduction

Zhaoxia Yu
Professor, Department of Statistics

2026-10-15

Section 1

Intro

Review: Lecture 01-06

- Lecture 01: LSE in simple linear regression (SLR)
- Lectures 02-04: statistical inference in SLR, t-test, F-test, ANOVA.
- Lecture 05: Diagnostics in SLR.
- Lecture 06: data transformation and lack of fit test.
- Today: Multiple Linear Regression (MLR) - Introduction.

Section 2

MLR

SLR vs MLR

- SLR: In a simple linear regression (SLR) model, we have one response variable y and one predictor variable x.

$$y = \beta_0 + \beta_1 x + \epsilon$$

  - For individual i, $i = 1, \ldots, n$,

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

- MLR: In a multiple linear regression (MLR) model, we have one response variable y and multiple predictor variables $X_1, X_2, \ldots, X_p$.

$$y = \beta_0 + \beta_1 x_1 + \cdots, \beta_p x_p + \epsilon$$

MLR

- For indivuidual i, $i = 1, \ldots, n$,

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i$$

  where

- y_i is the response variable for individual i.
- x_{ij} is the value of predictor variable j for individual i.
- ϵ_i's are errors:
  - The assumptions on ϵ_i's are the same as those in SLR.
  - weaker version: ϵ_i's are independent with mean 0 and constant variance σ^2.
  - Strong version: ϵ_i's are i.i.d. $N(0, \sigma^2)$.
- The parameters to be estimated are $\beta_0, \beta_1, \ldots, \beta_p$ and σ^2.

Notations

- When putting all individuals together, we have

$$
\begin{aligned}
y_1 &= \beta_0 + \beta_1 x_{11} + \cdots + \beta_p x_{1p} + \epsilon_1 \\
y_2 &= \beta_0 + \beta_1 x_{21} + \cdots + \beta_p x_{2p} + \epsilon_2 \\
&\vdots \\
y_n &= \beta_0 + \beta_1 x_{n1} + \cdots + \beta_p x_{np} + \epsilon_n
\end{aligned}
$$

Matrix Form

- In matrix form, we have

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

Matrix Form

More compactly,

$$\mathbf{y} = \mathbf{X}\beta + \boldsymbol{\epsilon}$$

here

- $\mathbf{y}$ is the response vector of size $n \times 1$.
- $\mathbf{X}$ is the design matrix of size $n \times (p+1)$.
- β is the parameter vector of size $(p+1) \times 1$.
- $\boldsymbol{\epsilon}$ is the error vector of size $n \times 1$, where $E(\boldsymbol{\epsilon}) = \mathbf{0}$ and $Var(\boldsymbol{\epsilon}) = \sigma^2 \mathbf{I}_n$. Here $\sigma^2 \mathbf{I}_n$ is the variance covariance of $\boldsymbol{\epsilon}$ - it is a diagonal matrix with diagonal elements σ^2.

Matrix Form

- $\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$ is the response vector of size $n \times 1$.
- $\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$ is the design matrix of size $n \times (p+1)$.
- $\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}$ is the parameter vector of size $(p+1) \times 1$.
- $\boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{bmatrix}$ is the error vector of size $n \times 1$.

Section 3

Normal Equations

Least Squares Estimation (LSE) for MLR

- The least squares estimation (LSE) of β is obtained by minimizing the residual sum of squares (RSS):

$$RSS(\beta) = \sum_{i=1}^{n}(y_i - E[y_i])^2 = (\mathbf{y} - \mathbf{X}\beta)^T(\mathbf{y} - \mathbf{X}\beta)$$

where $E[y_i] = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$ is the fitted value for individual i.

LSE for MLR: $p = 2$

- Let's consider $p = 2$ for illustration:

$$RSS(\beta_0, \beta_1, \beta_2) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2})^2$$

- To find the minimum, we take the partial derivatives with respect to $\beta_0, \beta_1, \beta_2$ and set them to zero:

$$\frac{\partial RSS}{\partial \beta_0} = -2 \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2}) = 0$$

$$\frac{\partial RSS}{\partial \beta_1} = -2 \sum_{i=1}^{n} x_{i1} (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2}) = 0$$

$$\frac{\partial RSS}{\partial \beta_2} = -2 \sum_{i=1}^{n} x_{i2} (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2}) = 0$$

Normal Equations: $p = 2$

- Note that there are three linear equations. They can be rewritten as:

$$n\beta_0 + \beta_1 \sum_{i=1}^{n} x_{i1} + \beta_2 \sum_{i=1}^{n} x_{i2} = \sum_{i=1}^{n} y_i$$

$$\beta_0 \sum_{i=1}^{n} x_{i1} + \beta_1 \sum_{i=1}^{n} x_{i1}^2 + \beta_2 \sum_{i=1}^{n} x_{i1}x_{i2} = \sum_{i=1}^{n} x_{i1}y_i$$

$$\beta_0 \sum_{i=1}^{n} x_{i2} + \beta_1 \sum_{i=1}^{n} x_{i1}x_{i2} + \beta_2 \sum_{i=1}^{n} x_{i2}^2 = \sum_{i=1}^{n} x_{i2}y_i$$

- These equations are called the normal equations.

Normal Equations: $p = 2$

- To connect the normal equations to their matrix form, we can write them as:

$$\begin{bmatrix} n & \sum_{i=1}^n x_{i1} & \sum_{i=1}^n x_{i2} \\ \sum_{i=1}^n x_{i1} & \sum_{i=1}^n x_{i1}^2 & \sum_{i=1}^n x_{i1}x_{i2} \\ \sum_{i=1}^n x_{i2} & \sum_{i=1}^n x_{i1}x_{i2} & \sum_{i=1}^n x_{i2}^2 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n x_{i1}y_i \\ \sum_{i=1}^n x_{i2}y_i \end{bmatrix}$$

Normal Equations: Matrix Form for $p = 2$

Recall that

- $\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} \\ 1 & x_{21} & x_{22} \\ \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} \end{bmatrix}$ is the design matrix of size $n \times (p+1)$.
- $\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$ is the parameter vector of size $(p+1) \times 1$.
- $\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$ is the response vector of size $n \times 1$.
- We can verify that the normal equations can be written as:

$$\mathbf{X}^T \mathbf{X} \beta = \mathbf{X}^T \mathbf{y}$$

- How to solve for β?

Section 4

LSE and Fitted Values

Normal Equations: Matrix Form for general p

- For general p, the normal equations can be written as:

$$\mathbf{X}^T\mathbf{X}\beta = \mathbf{X}^T\mathbf{y}$$

- If $\mathbf{X}^T\mathbf{X}$ is invertible, the LSE of β is given by:

$$\hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_p \end{bmatrix} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

Matrix Form Deriviation of LSE

- The RSS can be expanded as:

$$\begin{aligned} RSS(\beta) &= (\mathbf{y} - \mathbf{X}\beta)^T(\mathbf{y} - \mathbf{X}\beta) \\ &= \mathbf{y}^T\mathbf{y} - 2\beta^T\mathbf{X}^T\mathbf{y} + \beta^T\mathbf{X}^T\mathbf{X}\beta \end{aligned}$$

- Taking the derivative with respect to β and setting it to zero, we have:

$$\frac{\partial RSS}{\partial \beta} = -2\mathbf{X}^T\mathbf{y} + 2\mathbf{X}^T\mathbf{X}\beta = 0$$

- This leads to the normal equations:

$$\mathbf{X}^T\mathbf{X}\beta = \mathbf{X}^T\mathbf{y}$$

MLR: fitted values

- The fitted values can be computed as:

$$\hat{\mathbf{y}} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \mathbf{X}\hat{\beta} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

- The hat matrix:

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$$

Note that $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$.

Section 5

Residuals

Residuals

- The residuals can be computed as:

$$\hat{\boldsymbol{\epsilon}} = \begin{bmatrix} \hat{\epsilon}_1 \\ \hat{\epsilon}_2 \\ \vdots \\ \hat{\epsilon}_n \end{bmatrix} = \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{X}\hat{\beta} = (\mathbf{I}_n - \mathbf{H})\mathbf{y}$$

SSE and MSE

- The sum of squared errors (SSE) can be computed as:

$$\begin{aligned} SSE &= \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \hat{\epsilon}^T\hat{\epsilon} = (\mathbf{y} - \mathbf{X}\hat{\beta})^T(\mathbf{y} - \mathbf{X}\hat{\beta}) \\ &= [(\mathbf{I}_n - \mathbf{H})\mathbf{y}]^T(\mathbf{I}_n - \mathbf{H})\mathbf{y} \\ &= \mathbf{y}^T(\mathbf{I}_n - \mathbf{H})\mathbf{y} \end{aligned}$$

- The mean squared error (MSE) is given by:

$$MSE = \frac{SSE}{n - (p+1)}$$

  where $n - (p+1)$ is the degrees of freedom (d.f.) for error.
- An unbiased estimator of σ^2 is given by $\hat{\sigma}^2 = MSE$.

Section 6

Example

Example: Heights

- This is a simulated data set to practice MLR.

```
set.seed(42); n <- 50
library(MASS)
parents <- mvrnorm(n,
  mu    = c(69.1, 63.7),
  Sigma = matrix(c(3^2, 0.2*3*2.8,
                   0.2*3*2.8, 2.8^2), 2, 2))
f.height <- parents[,1]; m.height <- parents[,2]

sex     <- sample(c("son","daughter"), n, TRUE)
sex_eff <- ifelse(sex=="son", 2.7, -2.7)
midp    <- (f.height + m.height)/2
c.height <- (69.1 - 0.60*((69.1+63.7)/2) - 2.7) +
  0.60*midp + sex_eff + rnorm(n, 0, 2.5)
height.data= data.frame(c.height, f.height, m.height)
```

Example: Heights

```
head(height.data)
```

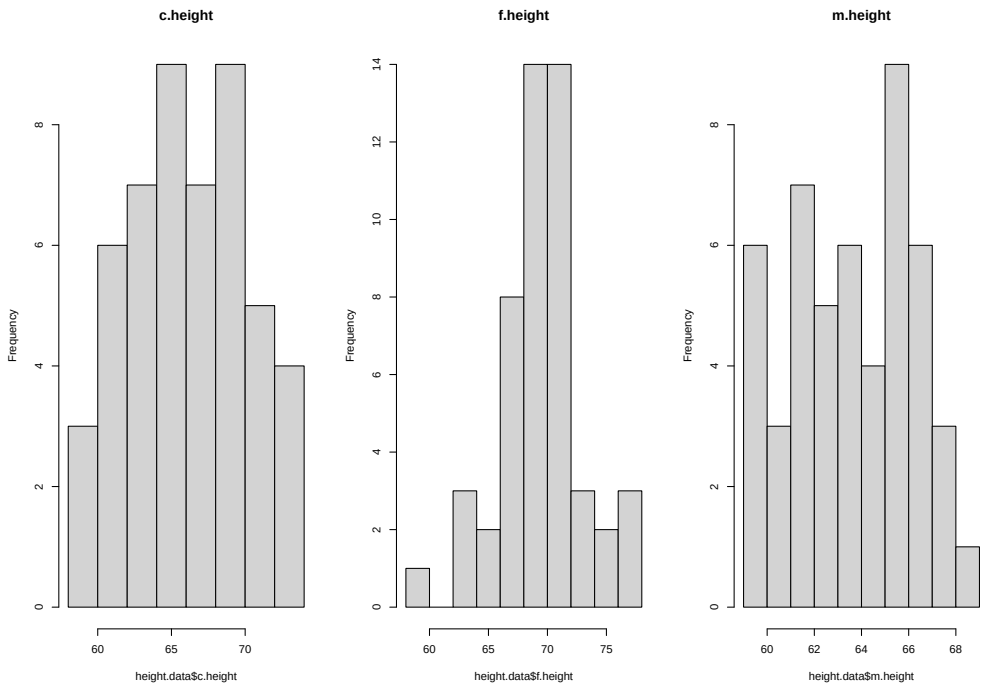
```
##   c.height f.height m.height
## 1 60.73932 66.01638 60.48351
## 2 63.06214 69.39601 66.39173
## 3 63.26804 70.51269 59.71977
## 4 59.67623 68.41591 61.17775
## 5 68.99239 68.18297 62.76237
## 6 70.61082 69.78964 63.31624
```

```
summary(height.data)
```

```
##     c.height        f.height        m.height
##  Min.   :58.63   Min.   :59.37   Min.   :59.07
##  1st Qu.:63.12   1st Qu.:66.97   1st Qu.:61.39
##  Median :66.52   Median :69.50   Median :63.49
##  Mean   :66.12   Mean   :69.34   Mean   :63.55
##  3rd Qu.:68.96   3rd Qu.:70.89   3rd Qu.:65.61
##  Max.   :73.89   Max.   :77.56   Max.   :68.57
```

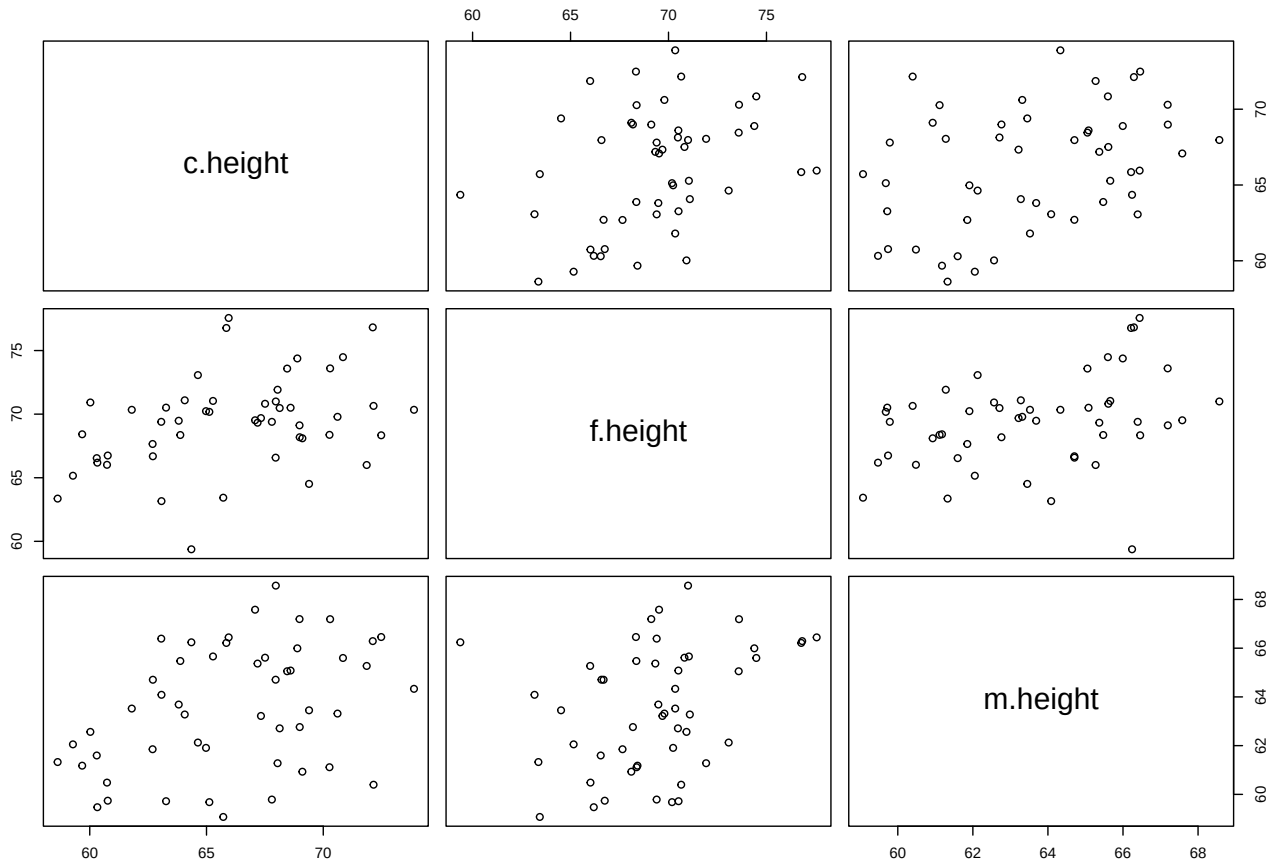
Example: Heights: Visualize data

```r
#Visualize data
par(mfrow=c(1,3))
hist(height.data$c.height, main="c.height")
hist(height.data$f.height, main="f.height")
hist(height.data$m.height, main="m.height")
```



Example: Heights: Visualize data

```
#Visualize data
pairs(height.data)
```



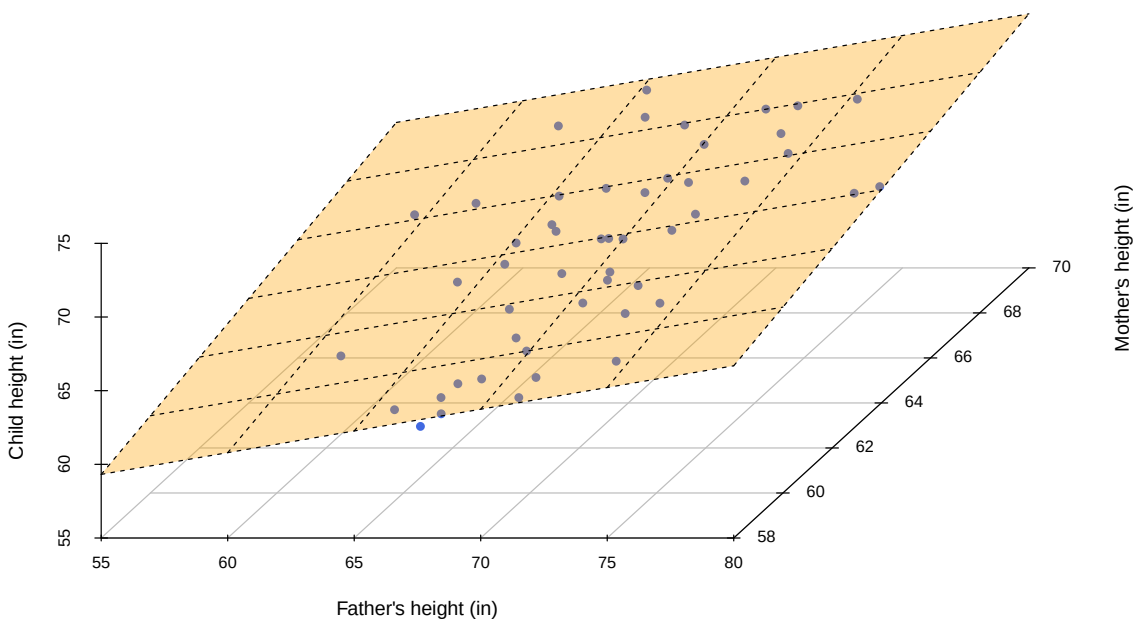
Example: Heights: MLR Results

```
height.lm= lm(c.height ~ f.height + m.height, data=height.data)
summary(height.lm)
```

```
##
## Call:
## lm(formula = c.height ~ f.height + m.height, data = height.data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.0987 -2.8324 -0.0332  2.2544  7.1131
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  16.2580    14.1585   1.148   0.2567
## f.height      0.2949     0.1508   1.955   0.0565 .
## m.height      0.4628     0.2148   2.155   0.0363 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.568 on 47 degrees of freedom
## Multiple R-squared:  0.2128, Adjusted R-squared:  0.1793
## F-statistic: 6.352 on 2 and 47 DF,  p-value: 0.003615
```

Example: Heights: Fitted Model

$$\hat{c.height} = 16.26 + 0.29 \cdot f.height + 0.46 \cdot m.height$$



Example: Heights: MLR Fitted Model

- The fitted model is:

$$\hat{c.height} = 16.26 + 0.29 \cdot f.height + 0.46 \cdot m.height$$

- Interpretation of coefficients:
  - For a one unit increase in father's height, the son's/daughter's height is expected to increase by 0.29 inches, holding mother's height constant.
  - For a one unit increase in mother's height, the son's/daughter's height is expected to increase by 0.46 inches, holding father's height constant.

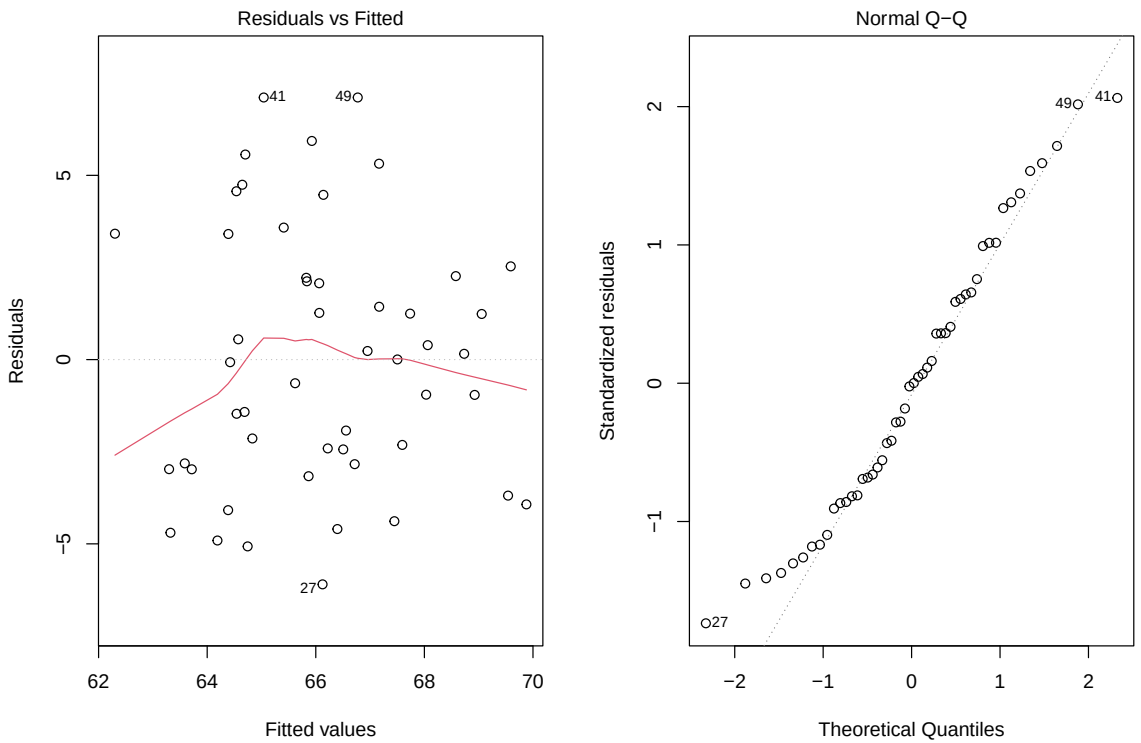
Example: Heights: MLR Results

From the summary,

- $n =$____, the number of observations.
- $R^2 =$____, the variance explained by the model (the two predictors, parents' heights).
- $s =$____, the estimated σ. Recall that $s^2 = MSE = SSE/(n - (p + 1))$.
- What else can you learn from the summary output?

Residual Plots

```
par(mfrow=c(1,2))
plot(height.lm, which=1:2)
```



Section 7

Inference

Statistical Inference About Regression Parameters

- Under the strong version of the model assumptions, we have:

$$\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^T\mathbf{X})^{-1})$$

- The variance (var-cov) of $\hat{\boldsymbol{\beta}}$ can be estimated by:

$$\widehat{Var}(\hat{\boldsymbol{\beta}}) = s^2(\mathbf{X}^T\mathbf{X})^{-1}$$

- The standard error (SE) of $\hat{\beta}_j$ is given by:

$$SE(\hat{\beta}_j) = s\sqrt{\cdot[(\mathbf{X}^T\mathbf{X})^{-1}]_{jj}}$$

where $[(\mathbf{X}^T\mathbf{X})^{-1}]_{jj}$ is the j-th diagonal element of the matrix $(\mathbf{X}^T\mathbf{X})^{-1}$.

t-tests For Regression Parameters

- To test the significance of each predictor variable, we can use the t-test:

$$t = \frac{\hat{\beta}_j - \beta_{j0}}{SE(\hat{\beta}_j)}$$

  where β_{j0} is the hypothesized value of β_j under the null hypothesis.
- Under the null hypothesis $H_0 : \beta_j = \beta_{j0}$, the test statistic follows a t-distribution with $n - (p + 1)$ degrees of freedom:

$$t \sim t_{n-(p+1)}$$

t-tests For Regression Parameters

- We mentioned that $\hat{\beta}_j$ follows a normal distribution. To construct a t statistic, we also need the following.
- It can be shown that $\hat{\beta}_j$ and s^2 are independent, and

$$\frac{(n-(p+1))s^2}{\sigma^2} = \frac{SSE}{\sigma^2} \sim \chi^2_{n-(p+1)}$$

- Therefore, we have

$$\frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)} = \frac{\hat{\beta}_j - \beta_j}{\sqrt{\sigma^2[(\mathbf{X}^T\mathbf{X})^{-1}]_{jj}}} \Big/ \sqrt{\frac{(n-(p+1))s^2}{\sigma^2(n-(p+1))}} \sim t_{n-(p+1)}$$

Confidence Intervals For Regression Parameters

- A $(1-\alpha)100\%$ confidence interval for β_j is given by:

$$\hat{\beta}_j \pm t_{\alpha/2,n-(p+1)} \cdot SE(\hat{\beta}_j)$$

  where $t_{\alpha/2,n-(p+1)}$ is the $(1-\alpha/2)$ quantile of the t-distribution with $n-(p+1)$ degrees of freedom.

- This confidence is derived from the t-test statistic, as follows:

$$1-\alpha = P\left(-t_{\alpha/2,n-(p+1)} \leq \frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)} \leq t_{\alpha/2,n-(p+1)}\right)$$

$$= P\left(\hat{\beta}_j - t_{\alpha/2,n-(p+1)} SE(\hat{\beta}_j) \leq \beta_j \leq \hat{\beta}_j + t_{\alpha/2,n-(p+1)} SE(\hat{\beta}_j)\right)$$

More Inference Problems